INCIDENT RECOVERY REPORT

RUNBOOK AGENT

A hand-written runbook built from the dev incidents.

22 incidents · run `baseline-rules` · generated 2026-10-30 03:56 UTC

100%	25%	99	35
recovered, dev incidents	recovered, never-seen incidents	mean score, dev	mean score, never-seen

HOW TO READ THIS REPORT

Each incident was injected into a live five-service shop. The agent investigated and acted only through a guarded toolbox, then filed a root-cause report. An independent verifier sent real user traffic over a soak window and checked that each fault's root cause was actually gone. Score: 60 points for a verified recovery, 30 for naming the right root cause, 10 for touching nothing that was not broken. Each incident page ends with recommendations: the immediate fix and how to prevent a repeat.

SUMMARY

SPLIT	INCIDENTS	RECOVERED	DIAGNOSIS F1	MEAN SCORE	COLLATERAL
dev	10	100%	0.97	99.0	0
held-out	12	25%	0.35	35.4	0

MEND

INCIDENTS

INCIDENT	SPLIT	RESULT	SCORE	REPORTED ROOT CAUSE
dev-01-api-down	dev	RECOVERED	100	api/service_down
dev-02-postgres-down	dev	RECOVERED	100	postgres/service_down
dev-03-redis-down	dev	RECOVERED	100	redis/service_down
dev-04-db-password	dev	RECOVERED	100	api/bad_config_deploy
dev-05-gateway-upstream	dev	RECOVERED	100	gateway/bad_config_deploy
dev-06-disk-full	dev	RECOVERED	100	api/disk_full
dev-07-cpu-hog	dev	RECOVERED	100	api/cpu_exhaustion
dev-08-postgres-partition	dev	RECOVERED	100	postgres/network_partition
dev-09-multi-disk-password	dev	RECOVERED	90	api/bad_config_deploy
dev-10-multi-redis-gateway	dev	RECOVERED	100	redis/service_down; gateway/bad_config_deploy
ho-01-db-host-dns	held-out	NOT RECOVERED	10	none
ho-02-crashloop	held-out	NOT RECOVERED	10	none
ho-03-partial-tax-rate	held-out	NOT RECOVERED	10	none
ho-04-memory-leak	held-out	NOT RECOVERED	10	none
ho-05-db-lock	held-out	NOT RECOVERED	10	none
ho-06-conn-exhaustion	held-out	NOT RECOVERED	10	none
ho-07-redis-partition	held-out	RECOVERED	100	redis/network_partition
ho-08-rollback-trap	held-out	RECOVERED	100	api/bad_config_deploy
ho-09-lock-behind-gateway	held-out	NOT RECOVERED	30	gateway/bad_config_deploy
ho-10-cpu-and-redis-partition	held-out	RECOVERED	100	redis/network_partition; api/cpu_exhaustion
ho-11-conn-leak-and-dns	held-out	NOT RECOVERED	10	none
ho-12-triple	held-out	NOT RECOVERED	25	redis/service_down

DEV

dev-01-api-down

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 2003ms over the last few seconds.
  - GET /api/health: 1/1 failed, statuses [504] (<html>
<head><title>504 Gateway Time-out</title></head>
<body>
<center><h1>50)
  - GET /api/products: 1/1 failed, statuses [504] (<html>
<head><title>504 Gateway Time-out</title></head>
<body>
<center><h1>50)
  - GET /api/orders/recent: 1/1 failed, statuses [504] (<html>
<head><title>504 Gateway Time-out</title></head>
<body>
<center><h1>50)
  - POST /api/orders: 1/1 failed, statuses [504] (<html>
<head><title>504 Gateway Time-out</title></head>
<body>
<center><h1>50)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`service_stopped` on `api`: A core service process is stopped.
Root causes to name: `api/service_down`

RUNBOOK AGENT'S INCIDENT REPORT

`api/service_down`: api was exited
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 12 TOOL CALLS · 1 STATE CHANGES · 12.8 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	**ACT**	`start_service(service=api)`
3	1.1s	OBSERVE	`check_endpoints()`
4	1.1s	OBSERVE	`get_logs(service=gateway, since_seconds=120, grep=upstream)`
5	1.1s	OBSERVE	`get_recent_changes()`
6	1.1s	OBSERVE	`get_logs(service=api, since_seconds=120)`
7	1.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
8	1.3s	OBSERVE	`get_metrics(service=api)`
9	3.5s	OBSERVE	`list_services()`
10	3.6s	OBSERVE	`check_endpoints()`
11	3.7s	OBSERVE	`verify_recovery()`

MEND

#	TIME	TYPE	TOOL CALL
12	12.8s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'service_down', 'evidence': 'api was exited'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100% user traffic success	**46 ms** p95 latency	**0** collateral actions	**1.00** diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Start the api service (start_service), then confirm the endpoints recover.

Prevent a repeat

- Set a restart policy (restart: unless-stopped) so a stopped container comes back on its own.
- Alert on service state and container exits, not only on failing user traffic.
- Record who stops production services and require a change ticket for manual stops.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-02-postgres-down

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 40%, p95 8ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: [Errno -2]
Name does not resolve","di)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`service_stopped` on `postgres`: A core service process is stopped.
Root causes to name: `postgres/service_down`

RUNBOOK AGENT'S INCIDENT REPORT

`postgres/service_down`: postgres was exited
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 10 TOOL CALLS · 1 STATE CHANGES · 12.4 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`start_service(service=postgres)`
3	0.5s	OBSERVE	`check_endpoints()`
4	0.5s	OBSERVE	`get_logs(service=api, since_seconds=120)`
5	0.6s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	0.7s	OBSERVE	`get_metrics(service=api)`
7	2.9s	OBSERVE	`list_services()`
8	3.1s	OBSERVE	`check_endpoints()`
9	3.2s	OBSERVE	`verify_recovery()`
10	12.4s	FINAL	`submit_report(root_causes=[{'component': 'postgres', 'category': 'service_down', 'evidence': 'postgres was exited'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	51 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

MEND

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Start the postgres service (start_service), then confirm the endpoints recover.

Prevent a repeat

- Set a restart policy (restart: unless-stopped) so a stopped container comes back on its own.
- Alert on service state and container exits, not only on failing user traffic.
- Record who stops production services and require a change ticket for manual stops.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

MEND

DEV

dev-03-redis-down

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 80%, p95 29ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [200] ({"checks":{"cache":"error: Error -2 connecting to
redis:6379. Name does not reso)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`service_stopped` on `redis`: A core service process is stopped.
Root causes to name: `redis/service_down`

RUNBOOK AGENT'S INCIDENT REPORT

`redis/service_down`: redis was exited
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 5 TOOL CALLS · 1 STATE CHANGES · 9.7 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`start_service(service=redis)`
3	0.6s	OBSERVE	`check_endpoints()`
4	0.7s	OBSERVE	`verify_recovery()`
5	9.8s	FINAL	`submit_report(root_causes=[{'component': 'redis', 'category': 'service_down', 'evidence': 'redis was exited'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	48 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Start the redis service (start_service), then confirm the endpoints recover.

Prevent a repeat

- Set a restart policy (restart: unless-stopped) so a stopped container comes back on its own.
- Alert on service state and container exits, not only on failing user traffic.
- Record who stops production services and require a change ticket for manual stops.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-04-db-password

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 40%, p95 29ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: connection
failed: connection to serv)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`bad_db_password` on `api`: Config deploy with wrong DB password.
Root causes to name: `api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`api/bad_config_deploy`: db password authentication failed
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 10 TOOL CALLS · 1 STATE CHANGES · 23.1 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.4s	ACT	`rollback_config(service=api)`
5	11.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	11.3s	OBSERVE	`get_metrics(service=api)`
7	13.5s	OBSERVE	`list_services()`
8	13.7s	OBSERVE	`check_endpoints()`
9	13.8s	OBSERVE	`verify_recovery()`
10	23.1s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'bad_config_deploy', 'evidence': 'db password authentication failed'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100% user traffic success	**47 ms** p95 latency	**0** collateral actions	**1.00** diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Roll back the api config to the last version with the working database password.

Prevent a repeat

- Rotate credentials through a secrets manager in step with the database, never by editing app config.
- Run a pre-deploy check that the new config can authenticate to the database.
- Deploy config as a canary and roll back automatically when health checks fail.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-05-gateway-upstream

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 2ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/products: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/orders/recent: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - POST /api/orders: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`gateway_bad_upstream` on `gateway`: Gateway config deploy with wrong upstream port.

Red herrings planted: `benign_api_deploy`

Root causes to name: `gateway/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`gateway/bad_config_deploy`: upstream connection refused after gateway deploy

Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 12 TOOL CALLS · 1 STATE CHANGES · 12.3 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=gateway, since_seconds=120, grep=upstream)`
4	0.2s	OBSERVE	`get_recent_changes()`
5	0.2s	ACT	`rollback_config(service=gateway)`
6	0.4s	OBSERVE	`get_logs(service=api, since_seconds=120)`
7	0.4s	OBSERVE	`run_diagnostic(service=api, check=disk)`
8	0.6s	OBSERVE	`get_metrics(service=api)`
9	2.9s	OBSERVE	`list_services()`
10	3.1s	OBSERVE	`check_endpoints()`
11	3.1s	OBSERVE	`verify_recovery()`

MEND

#	TIME	TYPE	TOOL CALL
12	12.3s	FINAL	`submit_report(root_causes=[{'component': 'gateway', 'category': 'bad_config_deploy', 'evidence': 'upstream connection refused after gateway deploy'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100% user traffic success	**69 ms** p95 latency	**0** collateral actions	**1.00** diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Roll back the gateway config to the last version with the correct api upstream port (api:8000).

Prevent a repeat

- Run nginx -t and an upstream connectivity check as part of every gateway deploy.
- Keep upstream addresses in one shared service registry instead of hand-edited configs.
- Deploy gateway changes with automatic rollback when 5xx rates jump.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-06-disk-full

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 80%, p95 31ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [200] ({"checks":{"cache":"ok","database":"ok","disk":"error:
[Errno 28] No space left )
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`disk_full` on `api`: A runaway export fills the API data volume.
Root causes to name: `api/disk_full`

RUNBOOK AGENT'S INCIDENT REPORT

`api/disk_full`: /data at 100%
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 10 TOOL CALLS · 1 STATE CHANGES · 12.4 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	0.5s	ACT	`delete_file(service=api, path=/data/tmp/export-orders-20261029.csv.tmp)`
6	0.6s	OBSERVE	`get_metrics(service=api)`
7	2.8s	OBSERVE	`list_services()`
8	3.0s	OBSERVE	`check_endpoints()`
9	3.1s	OBSERVE	`verify_recovery()`
10	12.4s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'disk_full', 'evidence': '/data at 100%'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	41 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Delete the runaway temporary export under /data/tmp/ on api, then confirm /data is below 80%.

Prevent a repeat

- Alert when the data volume passes 80% so it is caught before writes fail.
- Write exports to object storage or a separate volume with a quota, not to the app's data volume.
- Clean up temporary files automatically (TTL) and cap export sizes.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-07-cpu-hog

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 100%, p95 1399ms over the last few seconds.
  - all probes passing but p95 latency is 1399ms
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`cpu_hog` on `api`: A runaway search reindex job saturates the API's CPU.
Root causes to name: `api/cpu_exhaustion`

RUNBOOK AGENT'S INCIDENT REPORT

`api/cpu_exhaustion`: python /app/tools/reindex_search.py hogging CPU
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 11 TOOL CALLS · 1 STATE CHANGES · 19.0 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	2.1s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	2.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	3.7s	OBSERVE	`get_metrics(service=api)`
6	7.2s	OBSERVE	`run_diagnostic(service=api, check=processes)`
7	8.9s	ACT	`kill_process(service=api, pid=23)`
8	9.6s	OBSERVE	`list_services()`
9	9.8s	OBSERVE	`check_endpoints()`
10	9.9s	OBSERVE	`verify_recovery()`
11	19.0s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'cpu_exhaustion', 'evidence': 'python /app/tools/reindex_search.py hogging CPU'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	55 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Stop the runaway reindex_search job on api (kill_process), then confirm latency returns to normal.

Prevent a repeat

- Run batch jobs like search reindexing on a worker or a schedule, never inside the serving api container.
- Set CPU limits per container so one job cannot starve request handling.
- Alert on sustained high CPU together with rising p95 latency.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-08-postgres-partition

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 40%, p95 8ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: [Errno -2]
Name does not resolve","di)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`network_partition` on `postgres`: A backend service is cut off from the backend network.
Root causes to name: `postgres/network_partition`

RUNBOOK AGENT'S INCIDENT REPORT

`postgres/network_partition`: postgres missing from backend
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 5 TOOL CALLS · 1 STATE CHANGES · 9.5 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.1s	ACT	`reconnect_network(service=postgres, network=backend)`
3	0.2s	OBSERVE	`check_endpoints()`
4	0.3s	OBSERVE	`verify_recovery()`
5	9.5s	FINAL	`submit_report(root_causes=[{'component': 'postgres', 'category': 'network_partition', 'evidence': 'postgres missing from backend'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	42 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Reconnect postgres to the backend network (reconnect_network), then confirm its dependents recover.

Prevent a repeat

- Manage network attachments only through docker compose or infrastructure code, never by hand.
- Add connectivity checks between dependent services to monitoring.
- Alert on dependency health (database/cache unreachable) with the exact component named.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-09-multi-disk-password

SYSTEM RECOVERED

90 / 100

recovery 60 · diagnosis 20 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 36%, p95 30ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: connection
failed: connection to serv)
  - GET /api/products: 1/5 failed, statuses [200, 503] ({"error":"database unavailable"}
)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`disk_full` on `api`: A runaway export fills the API data volume.
`bad_db_password` on `api`: Config deploy with wrong DB password.
Root causes to name: `api/disk_full; api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`api/bad_config_deploy`: db password authentication failed
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 10 TOOL CALLS · 1 STATE CHANGES · 22.9 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	ACT	`rollback_config(service=api)`
5	11.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	11.4s	OBSERVE	`get_metrics(service=api)`
7	13.6s	OBSERVE	`list_services()`
8	13.8s	OBSERVE	`check_endpoints()`
9	13.8s	OBSERVE	`verify_recovery()`
10	22.9s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'bad_config_deploy', 'evidence': 'db password authentication failed'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	46 ms	0	0.67
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Delete the runaway temporary export under /data/tmp/ on api, then confirm /data is below 80%.
- Roll back the api config to the last version with the working database password.

Prevent a repeat

- Alert when the data volume passes 80% so it is caught before writes fail.
- Write exports to object storage or a separate volume with a quota, not to the app's data volume.
- Clean up temporary files automatically (TTL) and cap export sizes.
- Rotate credentials through a secrets manager in step with the database, never by editing app config.
- Run a pre-deploy check that the new config can authenticate to the database.
- Deploy config as a canary and roll back automatically when health checks fail.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

DEV

dev-10-multi-redis-gateway

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 2ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/products: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/orders/recent: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - POST /api/orders: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`service_stopped` on `redis`: A core service process is stopped.
`gateway_bad_upstream` on `gateway`: Gateway config deploy with wrong upstream port.
Red herrings planted: `worker_warnings`
Root causes to name: `redis/service_down; gateway/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`redis/service_down`: redis was exited
`gateway/bad_config_deploy`: upstream connection refused after gateway deploy
Summary: runbook applied 2 fix(es)

INVESTIGATION AND ACTIONS · 13 TOOL CALLS · 2 STATE CHANGES · 12.5 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`start_service(service=redis)`
3	0.6s	OBSERVE	`check_endpoints()`
4	0.6s	OBSERVE	`get_logs(service=gateway, since_seconds=120, grep=upstream)`
5	0.6s	OBSERVE	`get_recent_changes()`
6	0.6s	ACT	`rollback_config(service=gateway)`
7	0.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
8	0.8s	OBSERVE	`run_diagnostic(service=api, check=disk)`
9	0.9s	OBSERVE	`get_metrics(service=api)`
10	3.2s	OBSERVE	`list_services()`

MEND

#	TIME	TYPE	TOOL CALL
11	3.4s	OBSERVE	`check_endpoints()`
12	3.5s	OBSERVE	`verify_recovery()`
13	12.5s	FINAL	`submit_report(root_causes=[{'component': 'redis', 'category': 'service_down', 'evidence': 'redis was exited'}, {'component': 'gateway', 'category': 'bad_config_deploy', 'evidence': 'upstream connection refused after gateway deploy'}], summary=runbook applied 2 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100% user traffic success	**54 ms** p95 latency	**0** collateral actions	**1.00** diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Start the redis service (start_service), then confirm the endpoints recover.
- Roll back the gateway config to the last version with the correct api upstream port (api:8000).

Prevent a repeat

- Set a restart policy (restart: unless-stopped) so a stopped container comes back on its own.
- Alert on service state and container exits, not only on failing user traffic.
- Record who stops production services and require a change ticket for manual stops.
- Run nginx -t and an upstream connectivity check as part of every gateway deploy.
- Keep upstream addresses in one shared service registry instead of hand-edited configs.
- Deploy gateway changes with automatic rollback when 5xx rates jump.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-01-db-host-dns

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 40%, p95 9ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: [Errno -2]
Name does not resolve","di)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`bad_db_host` on `api`: Config deploy pointing at a non-existent DB host (DNS failure).
Root causes to name: `api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 17 TOOL CALLS · 0 STATE CHANGES · 15.3 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	0.4s	OBSERVE	`get_metrics(service=api)`
6	2.6s	OBSERVE	`list_services()`
7	2.8s	OBSERVE	`check_endpoints()`
8	2.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
9	2.8s	OBSERVE	`run_diagnostic(service=api, check=disk)`
10	3.0s	OBSERVE	`get_metrics(service=api)`
11	4.7s	OBSERVE	`list_services()`
12	4.8s	OBSERVE	`check_endpoints()`
13	4.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
14	4.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`
15	5.0s	OBSERVE	`get_metrics(service=api)`
16	6.7s	OBSERVE	`verify_recovery()`

MEND

#	TIME	TYPE	TOOL CALL
17	15.3s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

20% user traffic success	**7 ms** p95 latency	**0** collateral actions	**0.00** diagnosis F1

Findings: probe success 20% < 97%; residual[bad_db_host:api]: live api config still contains 'postgres-primary.internal'

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Roll back the api config to the last version that points at the reachable database host (postgres).

Prevent a repeat

- Validate that every hostname in a config change resolves from inside the target environment before deploying.
- Change database endpoints behind a stable DNS alias, not by editing each service's config.
- Roll back automatically when the dependency health check fails right after a deploy.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-02-crashloop

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 2002ms over the last few seconds.
  - GET /api/health: 3/3 failed, statuses [502, 504] (<html>
<head><title>504 Gateway Time-out</title></head>
<body>
<center><h1>50)
  - GET /api/products: 3/3 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/orders/recent: 3/3 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - POST /api/orders: 3/3 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`crashloop_config` on `api`: Config deploy that makes the API crash on boot.
Root causes to name: `api/bad_config_deploy or api/service_down`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 19 TOOL CALLS · 0 STATE CHANGES · 32.5 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=gateway, since_seconds=120, grep=upstream)`
4	0.2s	OBSERVE	`get_recent_changes()`
5	0.2s	OBSERVE	`get_logs(service=api, since_seconds=120)`
6	0.5s	OBSERVE	`run_diagnostic(service=api, check=disk)`
7	0.5s	OBSERVE	`get_metrics(service=api)`
8	2.7s	OBSERVE	`list_services()`
9	2.9s	OBSERVE	`check_endpoints()`
10	6.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
11	6.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`
12	7.1s	OBSERVE	`get_metrics(service=api)`

MEND

#	TIME	TYPE	TOOL CALL
13	9.3s	OBSERVE	`list_services()`
14	9.5s	OBSERVE	`check_endpoints()`
15	17.5s	OBSERVE	`get_logs(service=api, since_seconds=120)`
16	17.6s	OBSERVE	`run_diagnostic(service=api, check=disk)`
17	17.6s	OBSERVE	`get_metrics(service=api)`
18	19.8s	OBSERVE	`verify_recovery()`
19	32.5s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

20% user traffic success	**2004 ms** p95 latency	**0** collateral actions	**0.00** diagnosis F1

Findings: probe success 20% < 97%; probe p95 2004ms > 400ms; api is restarting; residual[crashloop_config:api]: live api config still contains 'auditing:'

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Roll back the api config to the last version that boots, then confirm the service stays up with no new restarts.

Prevent a repeat

- Validate config against a schema in CI so unknown or renamed sections are rejected before deploy.
- Alert on restart loops (restart count rising) separately from plain downtime.
- Make the app fail with a clear config error message instead of crashing without one.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-03-partial-tax-rate

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 80%, p95 31ms over the last few seconds.
  - POST /api/orders: 5/5 failed, statuses [500] ({"error":"pricing error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`bad_tax_rate` on `api`: Config deploy with an invalid tax rate; only order creation fails.
Root causes to name: `api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 17 TOOL CALLS · 0 STATE CHANGES · 15.6 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	0.4s	OBSERVE	`get_metrics(service=api)`
6	2.6s	OBSERVE	`list_services()`
7	2.8s	OBSERVE	`check_endpoints()`
8	2.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
9	2.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`
10	3.0s	OBSERVE	`get_metrics(service=api)`
11	4.7s	OBSERVE	`list_services()`
12	4.8s	OBSERVE	`check_endpoints()`
13	4.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
14	4.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`
15	5.0s	OBSERVE	`get_metrics(service=api)`
16	6.7s	OBSERVE	`verify_recovery()`
17	15.6s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

MEND

INDEPENDENT VERIFICATION

80% user traffic success	**30 ms** p95 latency	**0** collateral actions	**0.00** diagnosis F1

Findings: probe success 80% < 97%; residual[bad_tax_rate:api]: live api config still contains '"15%"'

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Roll back the api config to the last version with a numeric tax rate, then confirm order creation works.

Prevent a repeat

- Type-check business settings (tax rate must be a number between 0 and 1) in the config UI and in CI.
- Monitor each user journey separately: here only order creation failed while health checks looked fine.
- Require review for finance-owned config changes that reach production.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-04-memory-leak

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 100%, p95 90ms over the last few seconds.
  - all probes passing but p95 latency is 90ms
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`memory_leak` on `api`: Feature flag enabling an unbounded in-process cache (OOM loop).
Root causes to name: `api/bad_config_deploy or api/memory_exhaustion`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 9 TOOL CALLS · 0 STATE CHANGES · 13.4 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.3s	OBSERVE	`check_endpoints()`
3	0.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	1.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	1.4s	OBSERVE	`get_metrics(service=api)`
6	3.7s	OBSERVE	`list_services()`
7	3.8s	OBSERVE	`check_endpoints()`
8	4.0s	OBSERVE	`verify_recovery()`
9	13.3s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

91%	44 ms	0	0.00
user traffic success	p95 latency	collateral actions	diagnosis F1

Findings: probe success 91% < 97%; api restarted during soak (crash loop?); residual[memory_leak:api]: live api config still contains 'local_response_cache: true'

MEND

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Roll back the config change that enabled the unbounded local cache, then confirm memory stays flat and OOM kills stop.

Prevent a repeat

- Give every in-process cache a size limit and eviction policy.
- Roll out feature flags gradually, watching memory, before enabling them everywhere.
- Alert on OOM kills and on memory growth trends, not only on outages.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-05-db-lock

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 60%, p95 2026ms over the last few seconds.
  - GET /api/orders/recent: 1/1 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 1/1 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`db_lock` on `postgres`: A stuck migration holds an exclusive lock on the orders table.
Root causes to name: `postgres/db_lock_contention`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 17 TOOL CALLS · 0 STATE CHANGES · 30.5 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	4.2s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	4.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	4.4s	OBSERVE	`get_metrics(service=api)`
6	6.7s	OBSERVE	`list_services()`
7	6.9s	OBSERVE	`check_endpoints()`
8	10.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
9	11.0s	OBSERVE	`run_diagnostic(service=api, check=disk)`
10	11.1s	OBSERVE	`get_metrics(service=api)`
11	13.3s	OBSERVE	`list_services()`
12	13.5s	OBSERVE	`check_endpoints()`
13	17.6s	OBSERVE	`get_logs(service=api, since_seconds=120)`
14	17.6s	OBSERVE	`run_diagnostic(service=api, check=disk)`
15	17.8s	OBSERVE	`get_metrics(service=api)`
16	20.0s	OBSERVE	`verify_recovery()`
17	30.5s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

60%	2027 ms	0	0.00
user traffic success	p95 latency	collateral actions	diagnosis F1

Findings: probe success 60% < 97%; probe p95 2027ms > 400ms; residual[db_lock:postgres]: migration-runner sessions: 1

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Terminate the migration-runner session holding the exclusive lock on orders (terminate_db_sessions).

Prevent a repeat

- Set lock_timeout and statement_timeout for migrations so they fail instead of blocking traffic.
- Run schema migrations with online, non-blocking patterns during low-traffic windows.
- Alert on sessions blocked by locks for more than a few seconds.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-06-conn-exhaustion

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 40%, p95 32ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: connection
failed: connection to serv)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`conn_exhaustion` on `worker`: A backfill job in the worker leaks DB connections (and reconnects).
Root causes to name: `worker/db_connection_exhaustion` or `postgres/db_connection_exhaustion`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 17 TOOL CALLS · 0 STATE CHANGES · 16.1 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	0.4s	OBSERVE	`get_metrics(service=api)`
6	2.7s	OBSERVE	`list_services()`
7	2.9s	OBSERVE	`check_endpoints()`
8	2.9s	OBSERVE	`get_logs(service=api, since_seconds=120)`
9	3.0s	OBSERVE	`run_diagnostic(service=api, check=disk)`
10	3.1s	OBSERVE	`get_metrics(service=api)`
11	4.8s	OBSERVE	`list_services()`
12	5.0s	OBSERVE	`check_endpoints()`
13	5.1s	OBSERVE	`get_logs(service=api, since_seconds=120)`
14	5.2s	OBSERVE	`run_diagnostic(service=api, check=disk)`
15	5.3s	OBSERVE	`get_metrics(service=api)`
16	6.8s	OBSERVE	`verify_recovery()`

MEND

#	TIME	TYPE	TOOL CALL
17	16.1s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

20% user traffic success	**60 ms** p95 latency	**0** collateral actions	**0.00** diagnosis F1

Findings: probe success 20% < 97%; residual[conn_exhaustion:worker]: 'report_backfill' still running in worker; backfill connections=37

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Stop the leaking report_backfill job in the worker and terminate its database sessions.

Prevent a repeat

- Use a bounded connection pool in every job, and close connections in finally blocks.
- Give each application its own connection limit (per-role CONNECTION LIMIT or a pooler such as PgBouncer).
- Alert when database connections pass 80% of max_connections.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-07-redis-partition

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 80%, p95 37ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [200] ({"checks":{"cache":"error: Error -2 connecting to
redis:6379. Name does not reso)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`network_partition` on `redis`: A backend service is cut off from the backend network.
Root causes to name: `redis/network_partition`

RUNBOOK AGENT'S INCIDENT REPORT

`redis/network_partition`: redis missing from backend
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 5 TOOL CALLS · 1 STATE CHANGES · 9.5 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`reconnect_network(service=redis, network=backend)`
3	0.3s	OBSERVE	`check_endpoints()`
4	0.4s	OBSERVE	`verify_recovery()`
5	9.5s	FINAL	`submit_report(root_causes=[{'component': 'redis', 'category': 'network_partition', 'evidence': 'redis missing from backend'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	50 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Reconnect redis to the backend network (reconnect_network), then confirm its dependents recover.

Prevent a repeat

- Manage network attachments only through docker compose or infrastructure code, never by hand.
- Add connectivity checks between dependent services to monitoring.
- Alert on dependency health (database/cache unreachable) with the exact component named.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-08-rollback-trap

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 48ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: connection
failed: connection to serv)
  - GET /api/products: 5/5 failed, statuses [503] ({"error":"database unavailable"}
)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`bad_db_password` on `api`: Config deploy with wrong DB password.

Red herrings planted: `benign_api_deploy`

Root causes to name: `api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`api/bad_config_deploy`: db password authentication failed

Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 16 TOOL CALLS · 2 STATE CHANGES · 36.4 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	ACT	`rollback_config(service=api)`
5	11.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	11.3s	OBSERVE	`get_metrics(service=api)`
7	13.6s	OBSERVE	`list_services()`
8	13.7s	OBSERVE	`check_endpoints()`
9	13.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
10	13.8s	ACT	`rollback_config(service=api)`
11	24.5s	OBSERVE	`run_diagnostic(service=api, check=disk)`
12	24.8s	OBSERVE	`get_metrics(service=api)`
13	27.1s	OBSERVE	`list_services()`
14	27.2s	OBSERVE	`check_endpoints()`
15	27.3s	OBSERVE	`verify_recovery()`

MEND

#	TIME	TYPE	TOOL CALL
16	36.4s	FINAL	`submit_report(root_causes=[{'component': 'api', 'category': 'bad_config_deploy', 'evidence': 'db password authentication failed'}], summary=runbook applied 1 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100% user traffic success	**57 ms** p95 latency	**0** collateral actions	**1.00** diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Roll back the api config to the last version with the working database password.

Prevent a repeat

- Rotate credentials through a secrets manager in step with the database, never by editing app config.
- Run a pre-deploy check that the new config can authenticate to the database.
- Deploy config as a canary and roll back automatically when health checks fail.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-09-lock-behind-gateway

STILL BROKEN

30 / 100

recovery 0 · diagnosis 20 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 20%, p95 3ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/products: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - GET /api/orders/recent: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
  - POST /api/orders: 5/5 failed, statuses [502] (<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`db_lock` on `postgres`: A stuck migration holds an exclusive lock on the orders table.
`gateway_bad_upstream` on `gateway`: Gateway config deploy with wrong upstream port.
Root causes to name: `postgres/db_lock_contention; gateway/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`gateway/bad_config_deploy`: upstream connection refused after gateway deploy
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 20 TOOL CALLS · 1 STATE CHANGES · 26.6 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=gateway, since_seconds=120, grep=upstream)`
4	0.3s	OBSERVE	`get_recent_changes()`
5	0.3s	ACT	`rollback_config(service=gateway)`
6	0.4s	OBSERVE	`get_logs(service=api, since_seconds=120)`
7	0.4s	OBSERVE	`run_diagnostic(service=api, check=disk)`
8	0.6s	OBSERVE	`get_metrics(service=api)`
9	2.8s	OBSERVE	`list_services()`
10	3.0s	OBSERVE	`check_endpoints()`
11	7.0s	OBSERVE	`get_logs(service=api, since_seconds=120)`

MEND

#	TIME	TYPE	TOOL CALL
12	7.1s	OBSERVE	`run_diagnostic(service=api, check=disk)`
13	7.2s	OBSERVE	`get_metrics(service=api)`
14	9.4s	OBSERVE	`list_services()`
15	9.5s	OBSERVE	`check_endpoints()`
16	13.6s	OBSERVE	`get_logs(service=api, since_seconds=120)`
17	13.7s	OBSERVE	`run_diagnostic(service=api, check=disk)`
18	13.8s	OBSERVE	`get_metrics(service=api)`
19	16.0s	OBSERVE	`verify_recovery()`
20	26.6s	FINAL	`submit_report(root_causes=[{'component': 'gateway', 'category': 'bad_config_deploy', 'evidence': 'upstream connection refused after gateway deploy'}], summary=runbook applied 1 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

60%	2059 ms	0	0.67
user traffic success	p95 latency	collateral actions	diagnosis F1

Findings: probe success 60% < 97%; probe p95 2059ms > 400ms; residual[db_lock:postgres]: migration-runner sessions: 1

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Terminate the migration-runner session holding the exclusive lock on orders (terminate_db_sessions).
- Roll back the gateway config to the last version with the correct api upstream port (api:8000).

Prevent a repeat

- Set lock_timeout and statement_timeout for migrations so they fail instead of blocking traffic.
- Run schema migrations with online, non-blocking patterns during low-traffic windows.
- Alert on sessions blocked by locks for more than a few seconds.
- Run nginx -t and an upstream connectivity check as part of every gateway deploy.
- Keep upstream addresses in one shared service registry instead of hand-edited configs.
- Deploy gateway changes with automatic rollback when 5xx rates jump.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-10-cpu-and-redis-partition

SYSTEM RECOVERED

100 / 100

recovery 60 · diagnosis 30 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 80%, p95 1087ms over the last few seconds.
  - GET /api/health: 2/2 failed, statuses [200] ({"checks":{"cache":"error: Error -2 connecting to
redis:6379. Name does not reso)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`cpu_hog` on `api`: A runaway search reindex job saturates the API's CPU.
`network_partition` on `redis`: A backend service is cut off from the backend network.
Red herrings planted: `benign_gateway_deploy`
Root causes to name: `api/cpu_exhaustion; redis/network_partition`

RUNBOOK AGENT'S INCIDENT REPORT

`redis/network_partition`: redis missing from backend
`api/cpu_exhaustion`: python /app/tools/reindex_search.py hogging CPU
Summary: runbook applied 2 fix(es)

INVESTIGATION AND ACTIONS · 12 TOOL CALLS · 2 STATE CHANGES · 19.7 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`reconnect_network(service=redis, network=backend)`
3	0.3s	OBSERVE	`check_endpoints()`
4	2.6s	OBSERVE	`get_logs(service=api, since_seconds=120)`
5	2.6s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	4.1s	OBSERVE	`get_metrics(service=api)`
7	7.7s	OBSERVE	`run_diagnostic(service=api, check=processes)`
8	9.1s	ACT	`kill_process(service=api, pid=23)`
9	10.4s	OBSERVE	`list_services()`
10	10.5s	OBSERVE	`check_endpoints()`
11	10.6s	OBSERVE	`verify_recovery()`
12	19.7s	FINAL	`submit_report(root_causes=[{'component': 'redis', 'category': 'network_partition', 'evidence': 'redis missing from backend'}, {'component': 'api', 'category': 'cpu_exhaustion', 'evidence': 'python /app/tools/reindex_search.py hogging CPU'}], summary=runbook applied 2 fix(es), recovered=True)`

INDEPENDENT VERIFICATION

100%	46 ms	0	1.00
user traffic success	p95 latency	collateral actions	diagnosis F1

RECOMMENDATIONS

Immediate fix **DONE BY THE AGENT**

- Stop the runaway reindex_search job on api (kill_process), then confirm latency returns to normal.
- Reconnect redis to the backend network (reconnect_network), then confirm its dependents recover.

Prevent a repeat

- Run batch jobs like search reindexing on a worker or a schedule, never inside the serving api container.
- Set CPU limits per container so one job cannot starve request handling.
- Alert on sustained high CPU together with rising p95 latency.
- Manage network attachments only through docker compose or infrastructure code, never by hand.
- Add connectivity checks between dependent services to monitoring.
- Alert on dependency health (database/cache unreachable) with the exact component named.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-11-conn-leak-and-dns

STILL BROKEN

10 / 100

recovery 0 · diagnosis 0 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 36%, p95 10ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [503] ({"checks":{"cache":"ok","database":"error: [Errno -2]
Name does not resolve","di)
  - GET /api/products: 1/5 failed, statuses [200, 503] ({"error":"database unavailable"}
)
  - GET /api/orders/recent: 5/5 failed, statuses [503] ({"error":"database error"}
)
  - POST /api/orders: 5/5 failed, statuses [503] ({"error":"database error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`conn_exhaustion` on `worker`: A backfill job in the worker leaks DB connections (and reconnects).
`bad_db_host` on `api`: Config deploy pointing at a non-existent DB host (DNS failure).
Root causes to name: `worker/db_connection_exhaustion or postgres/db_connection_exhaustion; api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

No root cause reported.
Summary: runbook applied 0 fix(es)

INVESTIGATION AND ACTIONS · 17 TOOL CALLS · 0 STATE CHANGES · 15.4 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	OBSERVE	`check_endpoints()`
3	0.2s	OBSERVE	`get_logs(service=api, since_seconds=120)`
4	0.3s	OBSERVE	`run_diagnostic(service=api, check=disk)`
5	0.4s	OBSERVE	`get_metrics(service=api)`
6	2.6s	OBSERVE	`list_services()`
7	2.8s	OBSERVE	`check_endpoints()`
8	2.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
9	2.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`
10	3.0s	OBSERVE	`get_metrics(service=api)`
11	4.7s	OBSERVE	`list_services()`
12	4.8s	OBSERVE	`check_endpoints()`
13	4.8s	OBSERVE	`get_logs(service=api, since_seconds=120)`
14	4.9s	OBSERVE	`run_diagnostic(service=api, check=disk)`

#	TIME	TYPE	TOOL CALL
15	5.0s	OBSERVE	`get_metrics(service=api)`
16	6.7s	OBSERVE	`verify_recovery()`
17	15.4s	FINAL	`submit_report(root_causes=[], summary=runbook applied 0 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

20% user traffic success	**7 ms** p95 latency	**0** collateral actions	**0.00** diagnosis F1

Findings: probe success 20% < 97%; residual[conn_exhaustion:worker]: 'report_backfill' still running in worker; backfill connections=37; residual[bad_db_host:api]: live api config still contains 'postgres-primary.internal'

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Stop the leaking report_backfill job in the worker and terminate its database sessions.
- Roll back the api config to the last version that points at the reachable database host (postgres).

Prevent a repeat

- Use a bounded connection pool in every job, and close connections in finally blocks.
- Give each application its own connection limit (per-role CONNECTION LIMIT or a pooler such as PgBouncer).
- Alert when database connections pass 80% of max_connections.
- Validate that every hostname in a config change resolves from inside the target environment before deploying.
- Change database endpoints behind a stable DNS alias, not by editing each service's config.
- Roll back automatically when the dependency health check fails right after a deploy.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.

HELD-OUT

ho-12-triple

STILL BROKEN

25 / 100

recovery 0 · diagnosis 15 · safety 10

PAGER ALERT THE AGENT RECEIVED

```
PAGE [SEV2] ShopStack synthetic monitoring is failing.
Probe success 60%, p95 32ms over the last few seconds.
  - GET /api/health: 5/5 failed, statuses [200] ({"checks":{"cache":"error: Error -2 connecting to
redis:6379. Name does not reso)
  - POST /api/orders: 5/5 failed, statuses [500] ({"error":"pricing error"}
)
Investigate, restore service, verify recovery, then submit an incident report.
```

WHAT WAS ACTUALLY BROKEN (HIDDEN FROM THE AGENT)

`service_stopped` on `redis`: A core service process is stopped.
`disk_full` on `api`: A runaway export fills the API data volume.
`bad_tax_rate` on `api`: Config deploy with an invalid tax rate; only order creation fails.
Root causes to name: `redis/service_down; api/disk_full; api/bad_config_deploy`

RUNBOOK AGENT'S INCIDENT REPORT

`redis/service_down`: redis was exited
Summary: runbook applied 1 fix(es)

INVESTIGATION AND ACTIONS · 18 TOOL CALLS · 1 STATE CHANGES · 16.0 S

#	TIME	TYPE	TOOL CALL
1	0.0s	OBSERVE	`list_services()`
2	0.2s	ACT	`start_service(service=redis)`
3	0.6s	OBSERVE	`check_endpoints()`
4	0.7s	OBSERVE	`get_logs(service=api, since_seconds=120)`
5	0.7s	OBSERVE	`run_diagnostic(service=api, check=disk)`
6	0.8s	OBSERVE	`get_metrics(service=api)`
7	3.1s	OBSERVE	`list_services()`
8	3.3s	OBSERVE	`check_endpoints()`
9	3.3s	OBSERVE	`get_logs(service=api, since_seconds=120)`
10	3.4s	OBSERVE	`run_diagnostic(service=api, check=disk)`
11	3.5s	OBSERVE	`get_metrics(service=api)`
12	5.1s	OBSERVE	`list_services()`
13	5.3s	OBSERVE	`check_endpoints()`
14	5.4s	OBSERVE	`get_logs(service=api, since_seconds=120)`
15	5.4s	OBSERVE	`run_diagnostic(service=api, check=disk)`
16	5.6s	OBSERVE	`get_metrics(service=api)`

MEND

#	TIME	TYPE	TOOL CALL
17	7.1s	OBSERVE	`verify_recovery()`
18	16.0s	FINAL	`submit_report(root_causes=[{'component': 'redis', 'category': 'service_down', 'evidence': 'redis was exited'}], summary=runbook applied 1 fix(es), recovered=False)`

INDEPENDENT VERIFICATION

80% user traffic success	**31 ms** p95 latency	**0** collateral actions	**0.50** diagnosis F1

Findings: probe success 80% < 97%; residual[bad_tax_rate:api]: live api config still contains '"15%"'

RECOMMENDATIONS

Immediate fix **STILL NEEDED**

- Start the redis service (start_service), then confirm the endpoints recover.
- Delete the runaway temporary export under /data/tmp/ on api, then confirm /data is below 80%.
- Roll back the api config to the last version with a numeric tax rate, then confirm order creation works.

Prevent a repeat

- Set a restart policy (restart: unless-stopped) so a stopped container comes back on its own.
- Alert on service state and container exits, not only on failing user traffic.
- Record who stops production services and require a change ticket for manual stops.
- Alert when the data volume passes 80% so it is caught before writes fail.
- Write exports to object storage or a separate volume with a quota, not to the app's data volume.
- Clean up temporary files automatically (TTL) and cap export sizes.
- Type-check business settings (tax rate must be a number between 0 and 1) in the config UI and in CI.
- Monitor each user journey separately: here only order creation failed while health checks looked fine.
- Require review for finance-owned config changes that reach production.

Fix and prevention advice come from the remediation playbook for the injected fault types, not from the agent.